



Student Performance Analysis: Predicting Academic Success

A Data Science Approach to Identifying Key Performance Factors

Introduction & Problem Statement

Why This Project Matters:

- Early identification of at-risk students
- Understanding key performance factors
- Data-driven insights over subjective assessments

Project Goal:

Build a predictive model to identify at-risk students and determine the most influential factors affecting success.



Dataset Overview

Key Information:

- 5,000 student records
- 15 original + 6 new features
- Target: Class_Grade (0-3 scale)

Feature Categories:

- Demographic: Age, Gender, Ethnicity
- Academic: GPA, Study Time Weekly
- Attendance: Absence_Rate, Absence Impact
- Support: Parental Education, Parental_Support, Tutoring
- Extracurricular: Sports, Music, Volunteering

Data Cleaning Process

01

Drop StudentID

Removed irrelevant unique identifier.

03

Rename Columns

Improved clarity for analysis.

02

Verify No Missing Values

Ensured 0% missing data.

04

Check Duplicates & Outliers

Confirmed no duplicates or significant outliers.

Cleaning ensures reliable analysis and modeling from real-world data inconsistencies.

Feature Engineering

Creating new features captures meaningful relationships and provides more predictive signals than raw data.

Is_Child

Age < 18 (captures developmental differences)

Study_Efficiency

StudyTime/Absences
(better than raw study hours)

Absence_Rate_Category

Categorized as Low/Medium/High

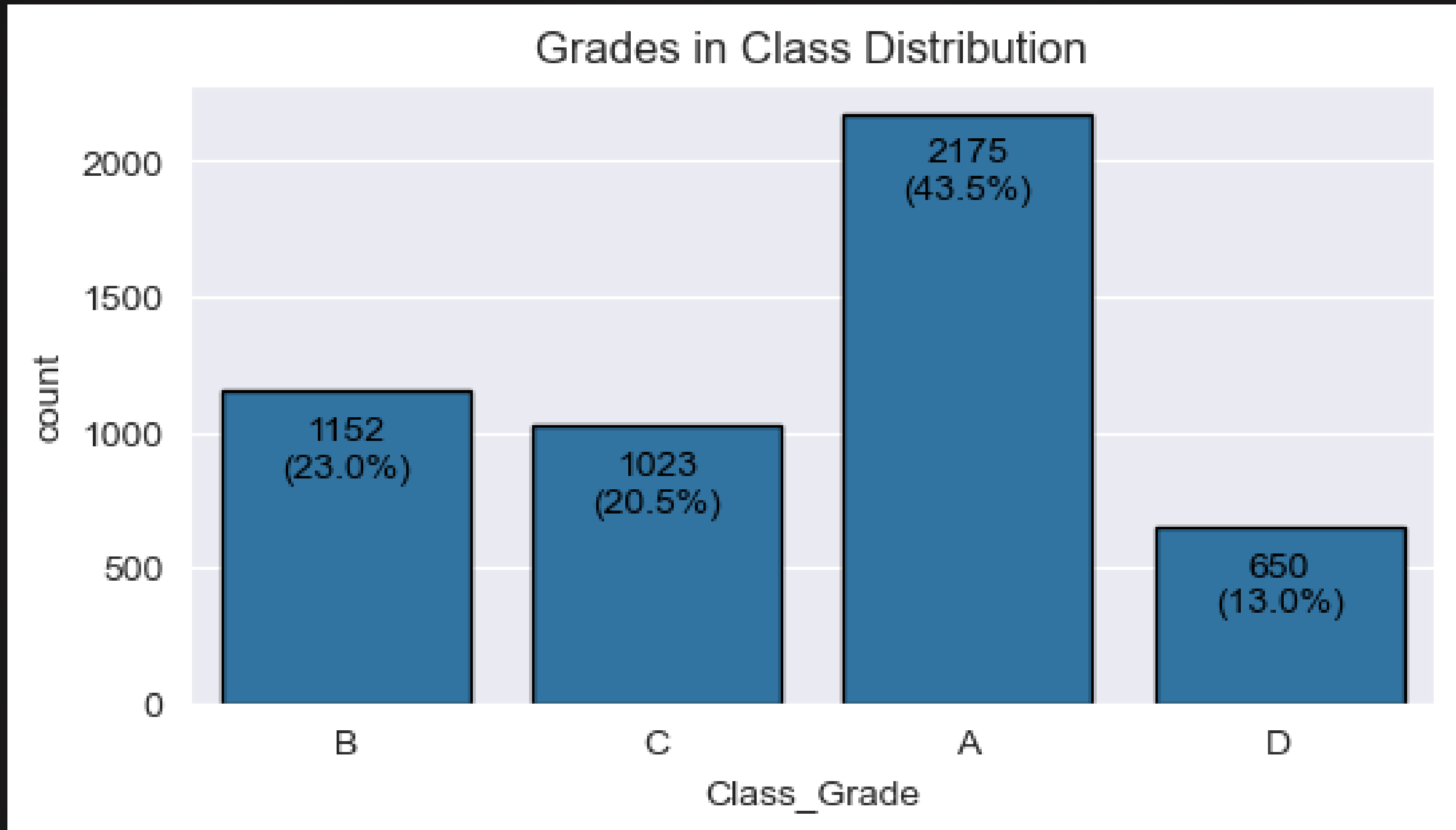
Total_Activities

Sum of extracurriculars

Absence_Impact & Study_Impact

Direct measures of effect on GPA

Target Variable Distribution

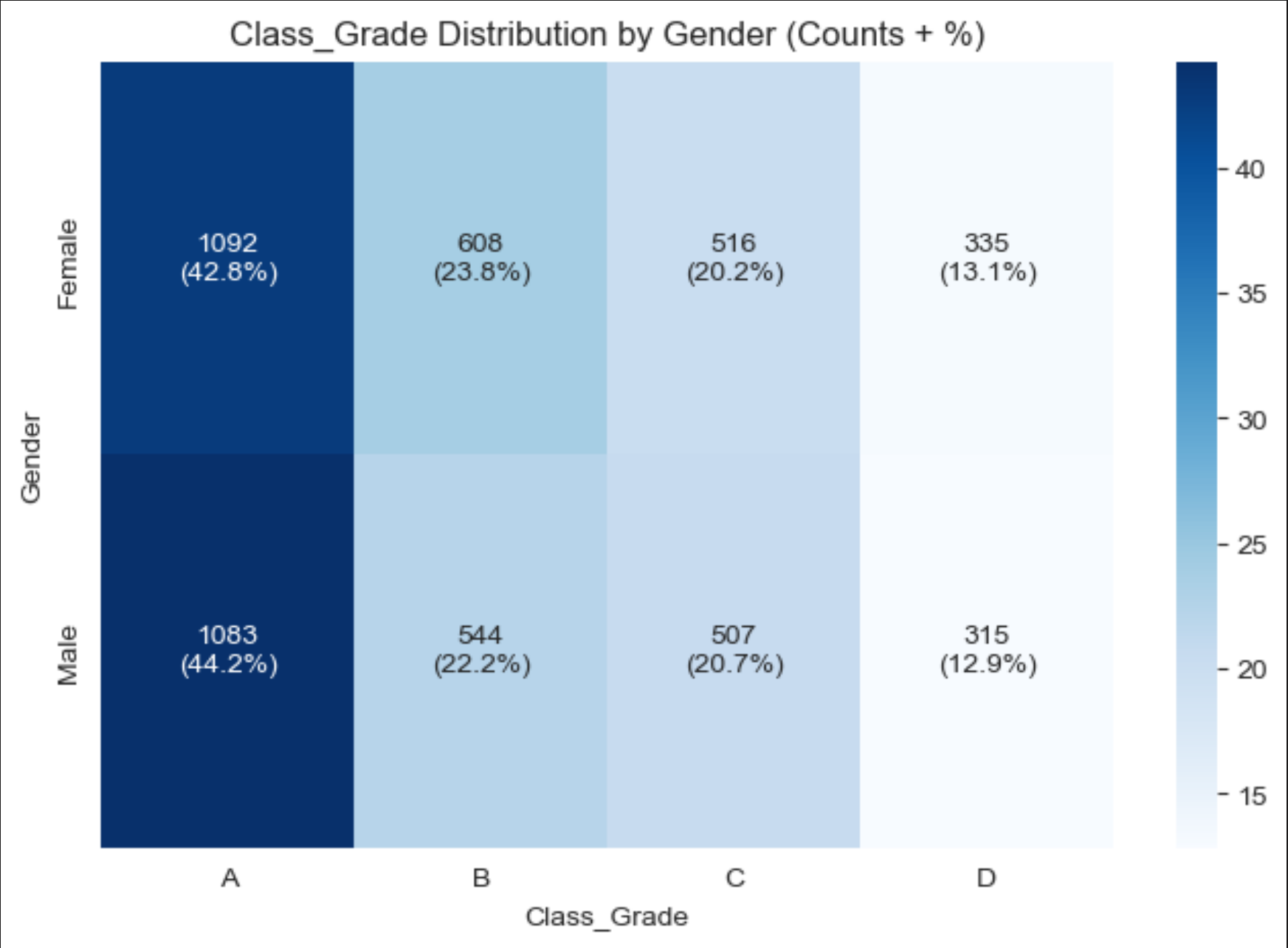


Gender vs. Academic Performance

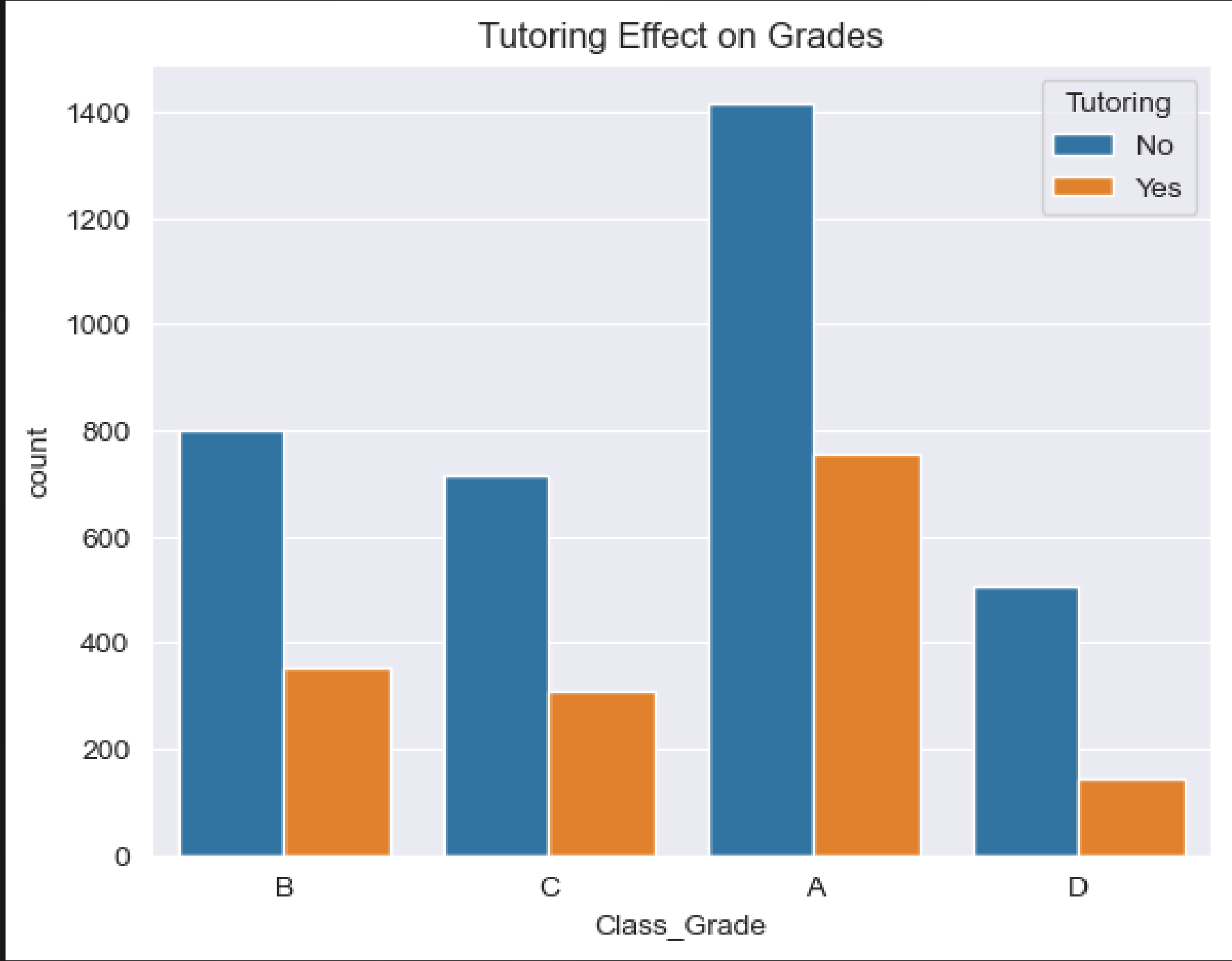
Female students consistently outperform male students across all grade categories, with the gap most pronounced in higher performance.

- 42.8% of female students achieved A grades.
- 44.2% of male students achieved A grades.

Recommendation: Investigate gender-specific learning approaches and consider tailored support for male students.



Tutoring Popularity



Tutoring has a clear positive effect on student performance

- Students with tutoring achieve a higher proportion of A Grades
- Tutoring significantly reduces the number of D Grades

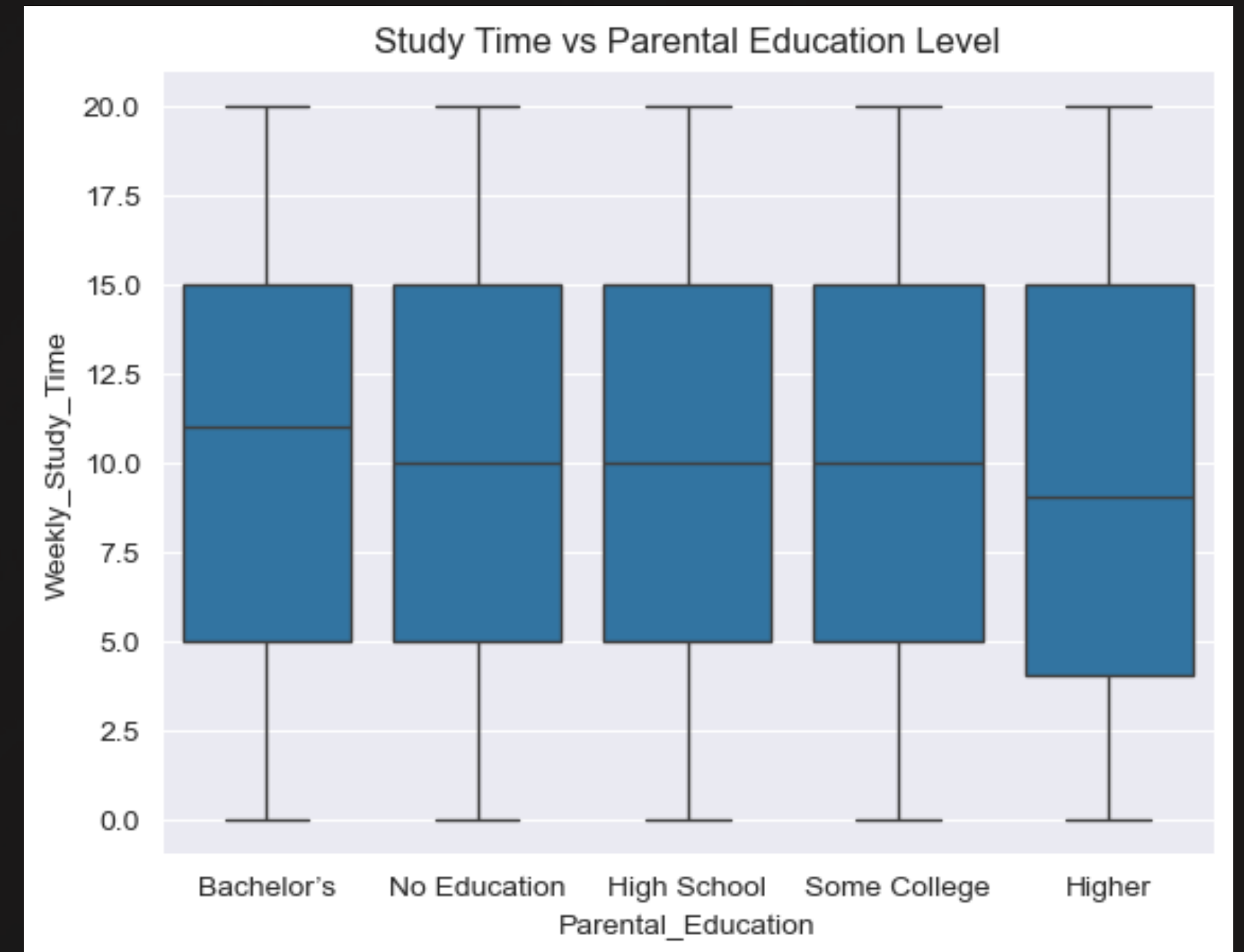
Recommendation: Encourage a wider access to tutoring programs to boost overall academic success and reduce failure rates.

Parental Education Impact

Parental education level shows little influence on students' weekly study time.

- Median study time remains similar across all parental education levels.
- Wide variation exists in every group, suggesting individual factors matter more.

Recommendation: Focus on motivating students directly rather than relying on parental background to influence study habits.



Unlocking Student Success: A Data- Driven Approach





Chapter 1

Key Insights from Student Data

Parental Education Impact

Higher parental education levels strongly correlate with better student performance.

Bachelor's+	35.7
High School or less	18.2

This clear progression highlights the need for additional support mechanisms for students from backgrounds with lower parental education.

Study Efficiency Analysis

Study efficiency (StudyTime/Absences) is a more critical factor than raw study hours for student success.

High Efficiency, Moderate Study

Outperformed those with high study time but poor attendance.

Optimal Balance

10-15 hours/week with fewer than 10 absences.

Diminishing Returns

Beyond 15 hours/week when attendance is good.

Focus interventions on improving attendance AND study habits, not just increasing study hours.



Chapter 2

Model Selection & Performance

Model Selection & Evaluation

We tested multiple machine learning models to ensure the most appropriate solution for predicting student performance.

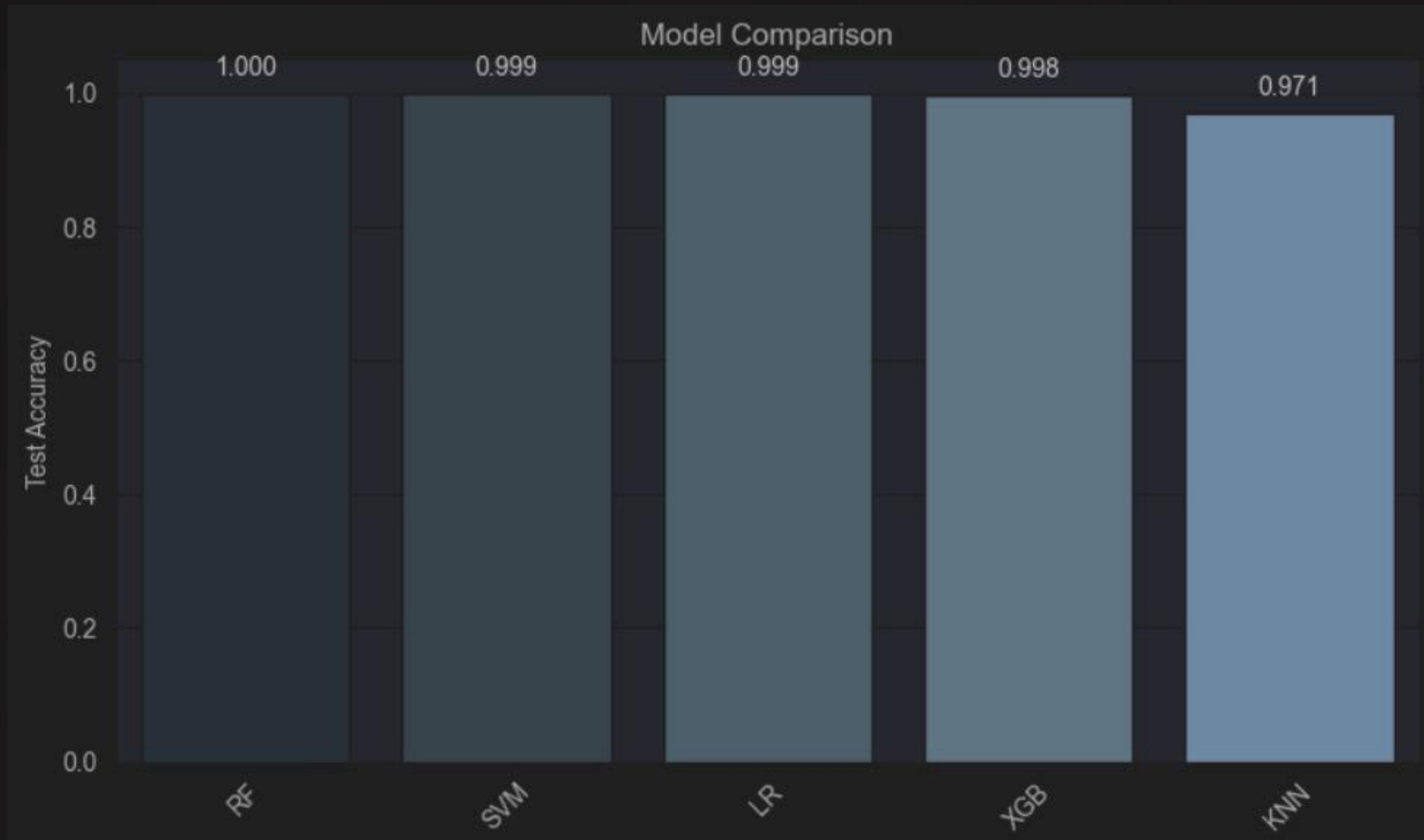
Models Tested

- Logistic Regression
- Support Vector Classifier (SVC)
- K-Nearest Neighbors (KNN)
- Random Forest
- XGBoost

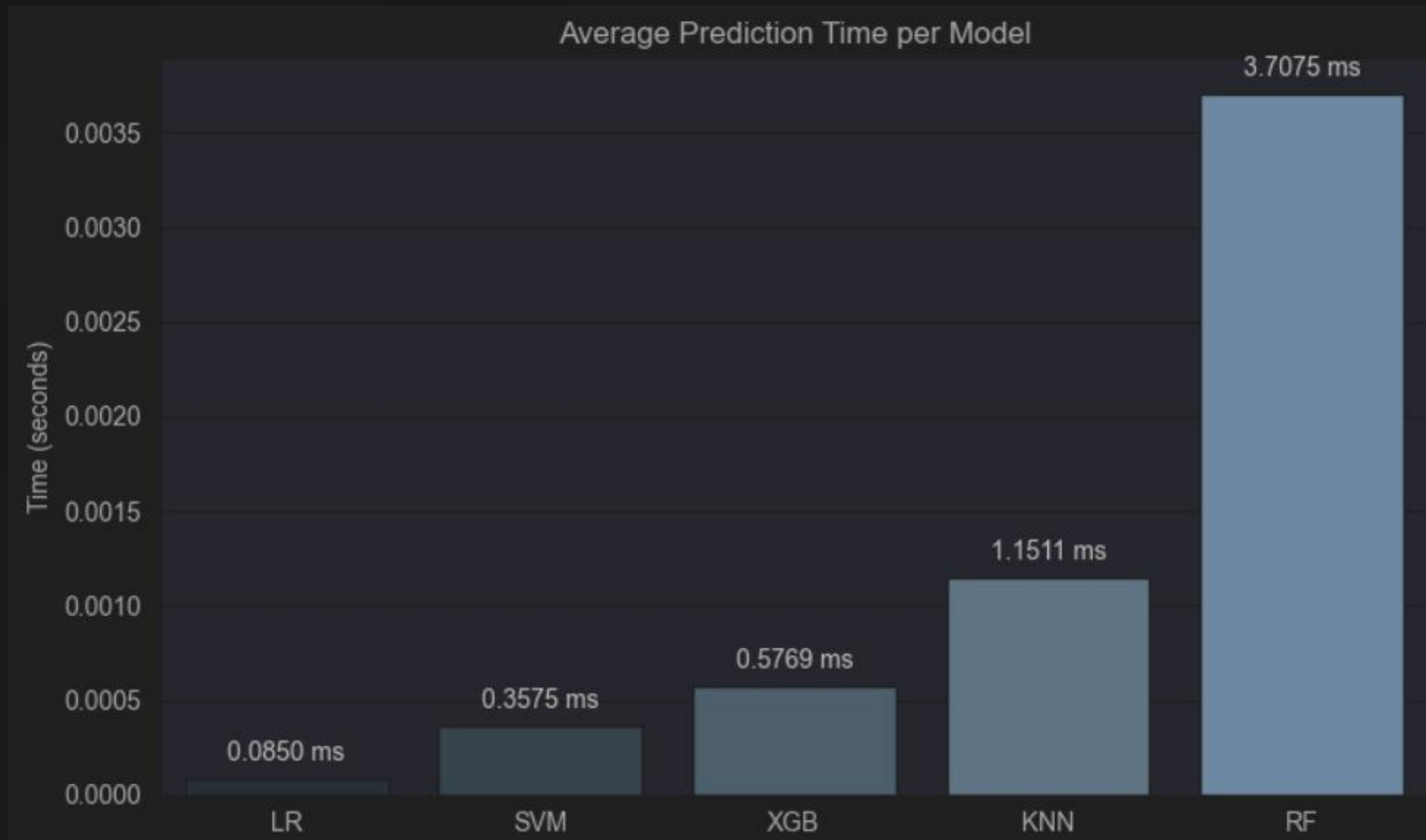
Evaluation Metrics

- Accuracy
- Log Loss
- Precision/Recall/F1
- Confusion Matrix analysis

Model Performance Comparison

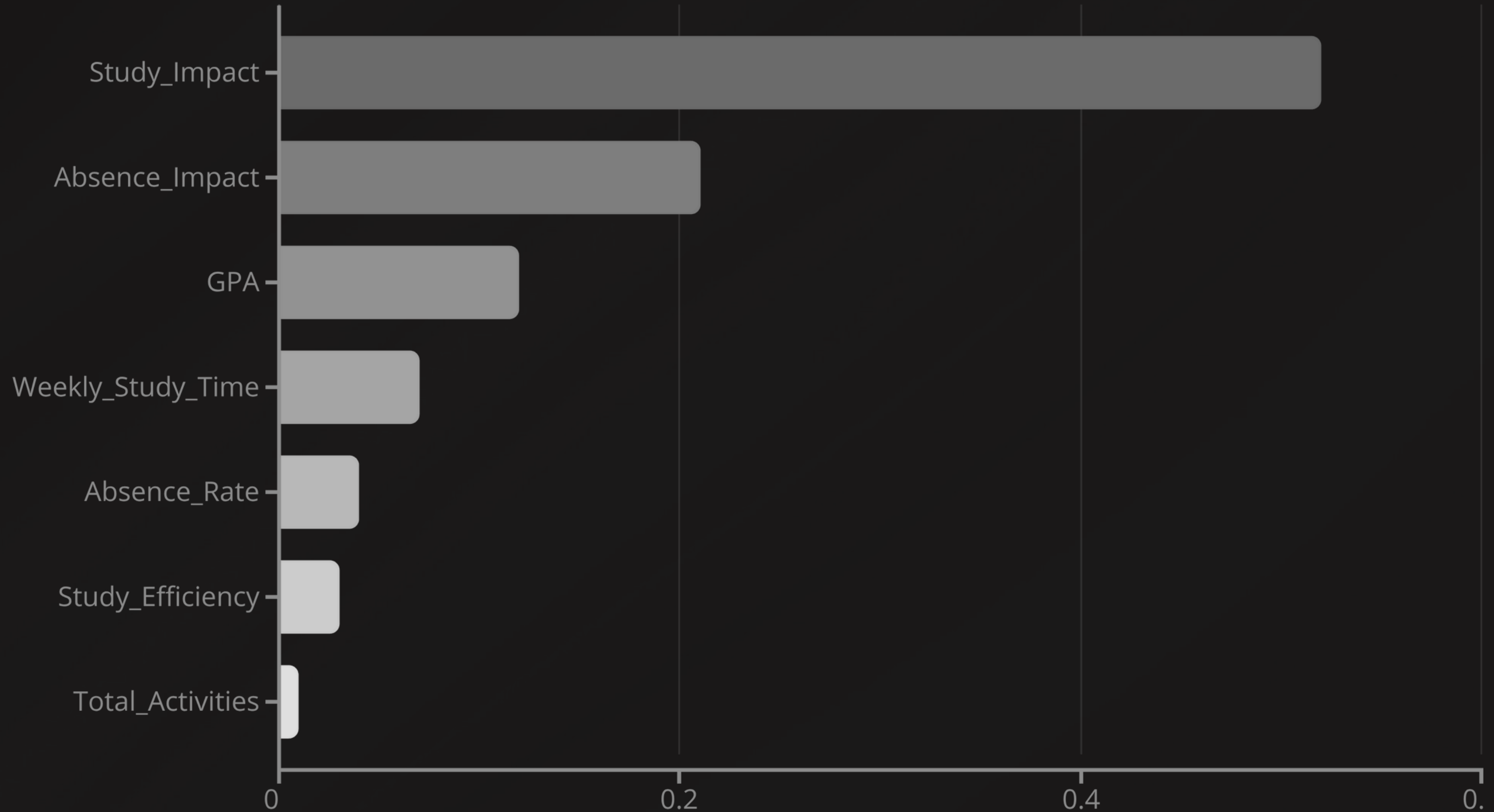


Model Performance Comparison



Feature Importance (Random Forest)

Our engineered impact features are significantly more predictive than raw features.



Key Insights Summary



Study Efficiency

Crucial for success, not just hours spent.



Attendance Impact

Stronger negative impact than study time's positive.



Parental Education

Low impact on student outcomes.



Targeted Tutoring

Most effective for struggling students.



Extracurriculars

Moderate involvement (1-2 activities) correlates with best outcomes.

These insights shift focus from traditional metrics to a more nuanced understanding of student performance drivers.

Actionable
Recommendations
For School
Administrators

01	02	03
Implement Early Warning System Track Absence_Rate and Study_Efficiency.	Target Tutoring For students with >10 absences or <5 hours weekly study.	Parental Engagement Programs Develop for families with lower education levels.
04	05	
Study Efficiency Workshops Focus on efficiency, not just more study hours.	Promote Balanced Extracurriculars Encourage 1-2 meaningful activities.	

Implementation Strategy: Start with pilot programs for at-risk students identified by our model before scaling.

Future Work Improvements

- Collect more granular study method data
- Add qualitative feedback mechanisms
- Track student performance over time
- Incorporate teacher assessments



Conclusion

Key Takeaways

- Student performance is multi-factorial - no single factor explains everything
- Our KNN model (97.1% accuracy) can reliably identify at-risk students
- Study efficiency and consistent attendance matter more than absolute study hours
- Targeted interventions based on data can significantly improve outcomes

Thank You

For more information or to discuss implementation, please contact us.

